INTERNHUB – AI-POWERED PERSONALIZED INTERNSHIP AND JOB RECOMMENDATION SYSTEM

A MINI PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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ABSTRACT

Internships and entry-level jobs give students practical knowledge, industry exposure and a start to their careers. Yet finding the right opportunity is hard. Students must search several platforms by hand, cannot tell which postings match their skills, and meet duplicate, outdated, fraudulent or incomplete listings along the way.

InternHub is an AI-powered web portal that gives personalised internship and job recommendations. The user uploads a resume, and the system analyses it with Natural Language Processing (NLP) to extract technical skills, programming languages, frameworks, tools, projects, certifications, domain and interests. To protect system integrity, a heuristic non-resume filter automatically flags and rejects financial documents such as tuition fee receipts or bills. InternHub then collects internships and jobs automatically from external APIs (including JobDataAPI, Jooble, Adzuna, Remotive, and Arbeitnow), converts them into one common format, checks them for missing information, purges dead listings, and removes duplicates using composite hashing.

Each opportunity is evaluated using a mathematically bounded Multi-Factor Hybrid Recommendation Engine combining Core Required Skills (50%), Preferred Skills (15%), Role Alignment (20%), Practical Project Evidence (10%), and Work-Mode/Location Preferences (5%). An automated Trust Score & Scam Indicator detects upfront fees and suspicious off-platform contact. For every recommended role, InternHub identifies missing competencies, provides actionable resume improvement suggestions, and generates a dynamic 4-phase learning roadmap.

Users can bookmark favourites and track applications across six statuses: Saved, Applied, Interview, Selected, Rejected, and Withdrawn. The application is built with React 19 and Vite on the frontend, Node.js and Express.js for API aggregation, Python with spaCy and scikit-learn for AI services, and PostgreSQL on Supabase with Row Level Security (RLS) for persistent storage and authentication. This report details the problem, literature review, architecture, database design, modules, algorithms, and comprehensive test validation.

Keywords: Internship and job recommendation, Resume analysis, Natural Language Processing, TF-IDF, Cosine similarity, Multi-factor scoring, Scam detection, Skill-gap analysis, Application tracking, Supabase.

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
AI	Artificial Intelligence
API	Application Programming Interface
CGPA	Cumulative Grade Point Average
CSS	Cascading Style Sheets
CRUD	Create, Read, Update, Delete
DFD	Data Flow Diagram
ER	Entity Relationship
FK	Foreign Key
HTML	HyperText Markup Language
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IDE	Integrated Development Environment
IDF	Inverse Document Frequency
IEEE	Institute of Electrical and Electronics Engineers
JS	JavaScript
JSON	JavaScript Object Notation
JWT	JSON Web Token
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
PDF	Portable Document Format
PK	Primary Key
RAM	Random Access Memory
REST	Representational State Transfer
RLS	Row Level Security
SQL	Structured Query Language

ABBREVIATION	EXPANSION
SSD	Solid State Drive
TF-IDF	Term Frequency–Inverse Document Frequency
TTL	Time To Live
UI / UX	User Interface / User Experience
URL	Uniform Resource Locator
UUID	Universally Unique Identifier

CHAPTER 1
INTRODUCTION

1.1 BACKGROUND

An internship is often a student's first real step from the classroom into the working world. It gives practical knowledge, a look at how an industry actually runs, and a chance to try a career before committing to it. After graduation the same search continues for entry-level jobs and graduate roles. For this reason almost every student in a technical course spends a good deal of time looking for opportunities.

The difficulty is not a shortage of openings. Hundreds of internships and jobs are posted every day on job portals, company websites and social platforms. The real difficulty is finding the few that suit one particular person. Most platforms show the same long list to everybody, whatever that person has studied, built or cares about. A student who is strong in Python and data analysis has to scroll past postings for accounting, sales and design before reaching the ones that matter.

Artificial Intelligence (AI) and Natural Language Processing (NLP) can help here. A computer can read a resume in a few seconds, understand which skills it mentions, and compare them with hundreds of job descriptions. InternHub is built on this idea.

1.2 PROBLEM STATEMENT

Students and job seekers often struggle to find suitable internships and jobs because they have to search several platforms by hand. They are also unsure which opportunities really match their skills, interests and career domain. Duplicate, outdated or incomplete listings make the search even harder. Table 1.1 summarises the main problems.

Table 1.1: Main problems faced by students and job seekers

Problem	What happens today	Effect on the student
Manual searching	The student visits many portals one by one and repeats the same search on each.	A lot of time is spent before a single application is written.
Unclear fit	Postings are not compared with the student's actual skills or interests.	The student cannot tell which opportunities are worth applying for.
Duplicate listings	The same job appears on several portals under slightly different details.	The same posting is read again and again.
Outdated listings	Some postings have expired or are missing the company, description or apply link.	Effort is wasted on openings that are closed or invalid.
Unknown skill gaps	Nobody tells the student which required skills are missing from the resume.	The student does not know what to learn next.
No tracking	Applications are noted in notebooks or spreadsheets, if at all.	Statuses and follow-ups are easily forgotten.

There is therefore a clear need for one system that reads a person's resume, collects current opportunities automatically, cleans the data, and recommends only the relevant ones.

1.3 MOTIVATION

As students ourselves, we have seen how much time goes into the search before a single application is even written. Much of that time is spent on work a computer can do better: comparing a long list of requirements against a resume, noticing that two postings are really the same job, or checking whether a listing has all the details needed to apply. We also noticed that many first-time applicants have no idea which skills they are missing. We wanted to build something that takes over this routine work, so that students can spend their time learning skills and preparing for interviews instead.

1.4 OBJECTIVES

The main objectives of the project, and how the system meets each of them, are given in Table 1.2.

Table 1.2: Objectives of the project

No.	Objective	How InternHub achieves it
1	Automate resume and skill analysis	Reads the uploaded resume with NLP and extracts skills, projects, certifications and domain.
2	Recommend suitable internships and jobs	Compares the student profile with every opportunity and ranks them by multi-factor match score.
3	Reduce manual searching across platforms	Collects listings from several external APIs into one place.
4	Provide personalised recommendations	Takes skills, domain, interests, location and work-mode preference into account.
5	Identify skill gaps	Compares the skills a role needs with the skills in the profile and lists what is missing.
6	Suggest resume and skill improvements	Gives specific tips based on the resume, target roles and the identified gaps.
7	Provide application tracking	Lets the student save opportunities and follow each application through its stages.
8	Keep opportunity data up to date	Fetches new listings periodically, updates changed ones and removes outdated ones.

1.5 PROJECT OVERVIEW

InternHub is an AI-powered web portal that gives personalised internship and job recommendations. The user creates an account, fills in a basic profile and uploads a resume. The system reads the resume, identifies the skills, domain and interests, and builds a structured profile.

In the background, InternHub collects current internships and jobs from external APIs. The data from different sources is converted into one common format, checked for missing information, and cleaned of duplicates. The remaining listings are compared with the user's profile and ranked by a match score. For each recommended opportunity the user can see the skills that are missing, receive suggestions for improving the resume, save the opportunity as a favourite, and track the application until the final outcome.

1.6 PROJECT DOMAIN

The primary domain of the project is Artificial Intelligence and Machine Learning. It also draws on the related areas listed in Table 1.3.

Table 1.3: Domains related to the project

Area	Role in InternHub
Natural Language Processing (NLP)	Reads resumes and job descriptions and extracts skills and domain.
Recommendation systems	Ranks opportunities for each user using content-based filtering.
Full-stack web development	Builds the user interface, the server logic and the database.
API integration	Collects live opportunities from external sources through REST APIs.
Data processing	Normalises, validates and de-duplicates the collected data.
Information retrieval	Finds and ranks the most relevant listings using TF-IDF and cosine similarity.

1.7 ORGANIZATION OF THE REPORT

The report is organised into thirteen chapters, described in Table 1.4.

Table 1.4: Organization of the report

Chapter	Title	What it covers
1	Introduction	Background, problem, motivation, objectives and domain.
2	Literature Survey	Five related research papers and the research gap.
3	Existing System	How people search today and the limits of that approach.
4	Proposed System	The InternHub solution, its features and how it differs from existing work.
5	System Requirements	Hardware, software, functional and non-functional requirements.
6	System Design	Architecture, workflow, API routes and database design.
7	Modules	The main modules and what each one does.
8	Algorithms and Techniques	NLP, TF-IDF, cosine similarity, duplicate detection and skill-gap logic.
9	Implementation	Technology stack and how each layer is built and secured.
10	Testing	Testing strategy and planned test cases.
11	Results & Discussion	Interface outputs, performance benchmarks and user benefits.
12	Conclusion	Summary of the work.
13	Future Enhancement	Directions for extending the project.

CHAPTER 2
LITERATURE SURVEY

2.1 OVERVIEW

Before designing InternHub, we studied five recent research papers on internship matching and recommendation. The aim was to learn which techniques other researchers have used, what worked well, and where gaps remain. Each paper is described in a table below, followed by a summary and the research gap that InternHub tries to fill.

2.2 PAPER 1: AI-BASED INTERNSHIP RECOMMENDATION SYSTEM

Table 2.1: Details of Paper 1

Title	AI-Based Internship Recommendation System
Authors	P. Madhavi Reddy, Geethanjali B. S., Shivang Karol, Vaishakh Pavan Ganesh R., R. Jennie Bharathi
Publication	IEEE Xplore, Vol. 12, published 18 March 2024
Methodology	The student registers and enters academic details, skills and preferences. The system pre-processes this data and extracts features such as skills, education and domain. AI-based recommendation algorithms then compare the student's profile with the internship requirements using content-based filtering and semantic analysis.
Merits	Gives personalised recommendations based on skills and interests. Reduces the time needed to find a suitable internship and improves accuracy using AI techniques. Offers a better user experience through relevant suggestions.
Demerits	Performance depends on manual data quality. Lacks automated resume parsing and does not connect to live external APIs.

2.3 PAPER 2: TECHTRACK – AN INTERNSHIP MATCHING PLATFORM

Table 2.2: Details of Paper 2

Title	TechTrack: An Internship Matching Platform for Career Recommendation Using Machine Learning
Publication	2026 11th International Conference on Business and Industrial Research (ICBIR), IEEE. DOI: 10.1109/ICBIR69932.2026.11584730
Methodology	TechTrack uses machine learning to match students with suitable internships and careers based on coursework, academic history, and early technical specializations, providing data-driven career guidance.
Merits	Provides accurate and personalised recommendations based on students' skills. Enhances overall internship search and advising experience.
Demerits	Requires manual entry; evaluates against closed institutional datasets rather than dynamic live market listings.

2.4 PAPER 3: AI-DRIVEN INTERNSHIP MATCHING USING SKILL-BASED RECOMMENDATION

Table 2.3: Details of Paper 3

Title	AI-Driven Internship Matching Using Skill-Based Recommendation System
Authors	Soorya Mol S. (PG Scholar) and Merin Manoj (Assistant Professor), Amal Jyothi College of Engineering
Publication	Proceedings of the National Conference on Emerging Computer Applications (NCECA) 2026, Vol. 7, Issue 1
Methodology	Proposes a skill-based recommendation system that matches a student's skills and profile with suitable internships using normalized feature vectors.
Merits	Improves matching efficiency and helps students find suitable internships quickly based on skill relevance.
Demerits	Sensitive to non-standard spelling or informal syntax; lacks live API feeds and post-match application tracking.

2.5 PAPER 4: AI-BASED INTERNSHIP ENGINE FOR THE PM INTERNSHIP SCHEME

Table 2.4: Details of Paper 4

Title	AI-Based Internship Recommendation Engine for the PM Internship Scheme
Authors	D. Rosy Salomi Victoria, Gokulraj L., Hem Kumar R., S. Roselin Mary, Nithya Roseline S.
Publication	IEEE Conference Publication, 2026. DOI: 10.1109/ICESIC67389.2026.11496549
Methodology	Presents an AI-based engine that recommends internships under the national PM Internship Scheme by matching candidate profiles with institutional requirements at scale.
Merits	Automates candidate matching at a national scale, substantially reducing administrative shortlisting overhead.
Demerits	Tailored to governmental quotas; lacks fine-grained software engineering skill taxonomies and automated PDF resume parsing.

2.6 PAPER 5: SMART INTERNSHIP RECOMMENDER (IRJMETS, 2026)

Table 2.5: Details of Paper 5

Title	Smart Internship Recommender
Authors	Apoorva Busad, Gopal Ujjwal Shankar, Dhanush S., Doddabasvaraj Abbigeri, Nandish Patil, Darshana KM
Publication	International Research Journal of Modernization in Engineering Technology and Science (IRJMETS), Vol. 08, Issue 01, January 2026, pp. 2733–2739. e-ISSN: 2582-5208
Methodology	Developed using Java J2EE (Servlets, JSP, JDBC, MySQL). Implements a deterministic weighted scoring model combining skill relevance and academic CGPA: FinalScore = alpha * SkillScore + beta * AcademicScore.
Merits	High explainability and computational efficiency (O(N*M) linear complexity) without requiring black-box deep learning.
Demerits	Requires manual entry of comma-separated skill strings; lacks synonym alias normalization and live multi-source API integration.

2.7 SUMMARY OF THE SURVEY

Table 2.6: Summary and comparison of the literature survey

No.	Paper	Main idea	Common limitation
1	AI-Based Internship (IEEE, 2024)	Content-based filtering & semantic analysis	Manual profile entry; static data
2	TechTrack (ICBIR, 2026)	ML-based career matching	Needs manual questionnaires
3	AI-Driven Matching (NCECA, 2026)	Skill-based vector matching	Brittle to syntax variations
4	PM Internship Engine (IEEE, 2026)	National-scale profile matching	Tailored to institutional quotas
5	Smart Recommender (IRJMETS, 2026)	Deterministic weighted matching	Manual inputs; no live APIs

2.8 RESEARCH GAP

The papers agree on one point: matching a student's skills with internship requirements gives far better results than a plain keyword search. They also share critical weaknesses: each depends heavily on manual data entry or closed offline datasets, and they concentrate solely on the matching step itself.

Little attention is paid to the full practical journey: building profiles automatically from uploaded resumes, rejecting spurious non-resume documents (such as fee receipts), aggregating live opportunities across multiple APIs, purging dead and duplicate listings, detecting deceptive recruitment scams, diagnosing skill gaps with sequential learning roadmaps, and providing persistent lifecycle tracking. InternHub unites this complete journey into a single platform.

CHAPTER 3
EXISTING SYSTEM

3.1 OVERVIEW

At present, students rely on general job portals, company career pages, social networking sites and college notice boards to find internships and jobs. A typical user creates accounts on several portals, types in a few keywords, and then opens listing after listing to read the requirements. Some portals send email alerts, but these are usually based on a job title or location the user typed in, not on the actual content of the resume.

3.2 LIMITATIONS OF THE EXISTING SYSTEM

Table 3.1: Limitations of the existing system

Limitation	Description
No personalisation	Portals mostly show opportunities as simple lists. They do not consider the user's skills, interests or career goals, so many irrelevant postings appear.
Keyword-based matching	Most systems look for the exact words the user typed. A relevant opportunity with a different job title is easily missed.
Manual search across platforms	Users must search, filter and compare postings on each platform themselves, which is slow and tiring.
Cold-start and data sparsity	Traditional recommenders do not work well for new users with no history, and they adapt slowly when interests change.
Duplicate, outdated or incomplete listings	The same job may appear several times, and some listings lack key details or have already closed.
No scam / fraud protection	Aggregators do not verify whether postings demand upfront fees or unvetted messaging channels.
No skill-gap guidance	Users are not told which skills they lack for the roles they want.
No place to track applications	Applications are recorded in notebooks or spreadsheets, so statuses and follow-ups are often lost.

3.3 NEED FOR A NEW SYSTEM

These limitations show the need for a single system that understands the user first, collects current opportunities on its own, cleans and checks them, and then recommends only what truly

fits. InternHub is designed to fill this gap.

CHAPTER 4
PROPOSED SYSTEM

4.1 OVERVIEW

InternHub is an AI-powered web portal that analyses a user's resume and profile to understand their skills, domain, interests and strengths. It then collects available internships and jobs from external sources, validates and processes the data, and recommends the opportunities that best fit the user's profile.

The platform also shows the skills the user is missing, suggests how to improve the resume, lets the user save favourites, and keeps a history of applications with their current status. By bringing together AI, NLP and live data, InternHub offers one place for opportunity discovery and career guidance.

4.2 KEY FEATURES

Table 4.1: Key features of InternHub

Feature	Description
Secure sign-up and login	Registration and authentication using Supabase Authentication with Row Level Security (RLS).
Resume upload and analysis	PDF resumes are read automatically; skills, languages, frameworks, tools and projects are extracted.
Spurious document guardrail	Heuristic patterns automatically reject non-resume uploads such as tuition receipts and bills.
Domain and interest detection	The suitable career domain (e.g., Web Development, Data Science, AI/ML) is identified from the resume.
Multi-source collection	Listings are fetched automatically from several live APIs (JobDataAPI, Jooble, Adzuna, Remotive, Arbeitnow).
Validation and deduplication	Incomplete listings are checked, dead links purged, and cross-source duplicates eliminated.
Scam & Trust Score	Evaluates listings against fraudulent patterns and outputs an objective Trust Score (0–100%).
Multi-factor match score	Combines Core Skills (50%), Preferred Skills (15%), Role Alignment (20%), and Preferences (5%).
Skill-gap analysis & roadmaps	Missing skills are highlighted alongside 4-phase sequential technology learning roadmaps.
Application tracking	Tracks applications across Saved, Applied, Interview, Selected, Rejected, and Withdrawn.

4.3 ADVANTAGES OF THE PROPOSED SYSTEM

Table 4.2: Advantages of the proposed system

Advantage	Benefit to the user
One place for all sources	No need to visit several platforms and repeat the same search.
Meaning-based matching	Results depend on the user's skills and not only on the words typed in a search box.
Cleaner & verified data	Duplicates and scam postings are reduced, ensuring safer and faster applications.
Clear next steps	The user knows exactly which skills to learn and what to change in the resume.
Everything organised	Saved opportunities and applications are kept together with their history.
Modular and extensible	New job sources or matching algorithms can be integrated without redesigning the architecture.

4.4 EXISTING SYSTEM VERSUS PROPOSED SYSTEM

Table 4.3: Existing system compared with InternHub

Aspect	Existing system	InternHub (proposed)
Search basis	Keywords typed by the user	Skills, domain and interests extracted from the resume
Sources	One portal at a time	Several live APIs combined in one place
Results	Generic lists for everybody	Personalised, ranked by multi-factor match score
Skill understanding	Exact word matching	TF-IDF with cosine similarity & canonical alias normalizer
Duplicates & scams	Appear repeatedly; unvetted	Validated, deduplicated, and scored for safety
Skill guidance	None	Skill-gap analysis and dynamic roadmaps
Tracking	Manual, outside the platform	Built-in application tracking with six statuses

4.5 HOW INTERNHUB DIFFERS FROM EXISTING PROJECTS

Figure 4.1 shows the main conceptual difference: existing portals focus strictly on the keyword-matching step, whereas InternHub manages the candidate's complete recruitment journey from initial PDF upload to final placement outcome.

Existing Approach:
[Visit Many Portals] → [Type Keywords] → [Read Generic Lists] → [Manual Tracking]

InternHub Approach:
[Upload PDF] → [NLP Skill & Bill Extraction] → [Live Multi-API Aggregator] → [Deduplicate & Trust Score] → [Multi-Factor Matching] → [Skill-Gap Roadmap] → [6-Stage Tracker]

Figure 4.1: Existing approach compared with the InternHub workflow pipeline

Table 4.4: Feature comparison with existing work

Feature	General job portals	Surveyed papers [1]–[5]	InternHub
Automatic Profile from PDF	No; user types info	Manual entry in all 5 papers	**Yes (PDF NLP Parser)**
Spurious Bill Rejection	None	Not described	**Yes (Heuristic Filter)**
Skill Matching with Score	Mostly keyword queries	Yes (main focus of papers)	**Yes (Multi-Factor Formula)**
Live Multi-Source APIs	Own listings only	Static offline datasets	**Yes (5 Live APIs)**
Scam / Fraud Trust Score	Reactive reporting	Not addressed	**Yes (Trust Score 0–100)**
Skill Gap & Roadmap	No	Not described	**Yes (Visual Diagnostics)**
Built-in Application Tracker	Manual, outside portal	Not supported	**Yes (6 Status States)**

CHAPTER 5
SYSTEM REQUIREMENTS SPECIFICATION

5.1 HARDWARE REQUIREMENTS

Table 5.1: Hardware requirements

Component	Requirement
Processor	Intel Core i5 / AMD Ryzen 5 or above (4 Cores, 2.5 GHz)
RAM	8 GB minimum (16 GB recommended for concurrent services)
Storage	256 GB SSD with at least 10 GB free space
Network	High-speed internet connection (≥ 10 Mbps)
Display	Any screen that can run a modern web browser (1080p recommended)

5.2 SOFTWARE REQUIREMENTS

Table 5.2: Software requirements

Component	Requirement
Operating system	Windows 10 / 11 (64-bit) or Linux Ubuntu 22.04 LTS
Frontend framework	React.js (v19.0.0), Vite (v8.0.0), TypeScript, Tailwind CSS v4
Backend framework	Node.js (v22.0.0+ LTS), Express.js (v4.21.0)
AI / ML engine	Python (v3.11+), spaCy, scikit-learn
Resume text extractor	PyMuPDF (fitz) or pdfplumber
Database & Storage	PostgreSQL hosted on Supabase Cloud
Authentication	Supabase Auth (JWT Bearer tokens) with Row Level Security
External API integration	JobDataAPI, Jooble, Adzuna, Remotive, Arbeitnow
Web Browser	Google Chrome, Microsoft Edge, or Mozilla Firefox

5.3 FUNCTIONAL REQUIREMENTS

Table 5.3: Functional requirements

ID	Area	The system should be able to…
FR1	Accounts	Let a user register, log in securely and manage a profile with domain, interests, location and work mode.
FR2	Resume	Accept a resume in PDF format (up to 10 MB) and read text streams safely into memory.
FR3	Guardrail	Identify and reject spurious non-resume uploads (such as tuition fee receipts and financial bills).
FR4	Resume NLP	Extract skills, programming languages, frameworks, tools and projects, and normalize aliases to canonical names.
FR5	Aggregation	Fetch internships and jobs automatically from multiple live recruitment APIs.
FR6	Normalization	Convert every external API response into one standardized schema and strip HTML tags.
FR7	Deduplication	Check each listing for key details and remove duplicates across disparate sources using composite hashing.
FR8	Trust Scoring	Evaluate descriptions for scam keywords and compute an automated Trust Score (0–100%).
FR9	Matching	Calculate a multi-factor match score between the profile and each opportunity and rank the results.
FR10	Guidance	Show missing skills and generate a dynamic sequential learning roadmap for the candidate.
FR11	Tracking	Enable saving favourites and tracking applications across Saved, Applied, Interview, Selected, Rejected, Withdrawn.
FR12	Updates	Refresh opportunity data periodically via in-memory caching and mirror availability checks.

5.4 NON-FUNCTIONAL REQUIREMENTS

Table 5.4: Non-functional requirements

Requirement	Description
Usability	Clean, responsive single-page interface with accessible contrast and intuitive navigation on desktop and mobile.
Security	JWT session authentication, protected environment API keys, and PostgreSQL Row Level Security (RLS).
Performance	End-to-end recommendation rendering in under 2.5 seconds; in-memory 5-minute TTL caching for fast repeated queries.
Reliability	Graceful degradation: if one external API fails, other providers continue unaffected without throwing server crashes.
Maintainability	Decoupled frontend, backend aggregation, AI microservice, and database layers for modular updates.
Explainability	Every recommendation exposes an explicit mathematical breakdown of matched vs. missing skills.
Data Freshness	Live mirror sync purges dead or closed vacancies to ensure applicants apply only to active roles.

CHAPTER 6
SYSTEM DESIGN

6.1 SYSTEM ARCHITECTURE

InternHub follows a layered, service-based cloud architecture. The presentation tier runs in modern web browsers using React 19, TypeScript, and Vite. The backend API gateway (Node.js and Express.js) orchestrates multi-provider live API aggregation, data cleansing, deduplication, and trust scoring. A dedicated Python NLP microservice executes document extraction, section parsing, and role prediction. Relational persistence and security are managed by Supabase PostgreSQL protected by Row Level Security (RLS).

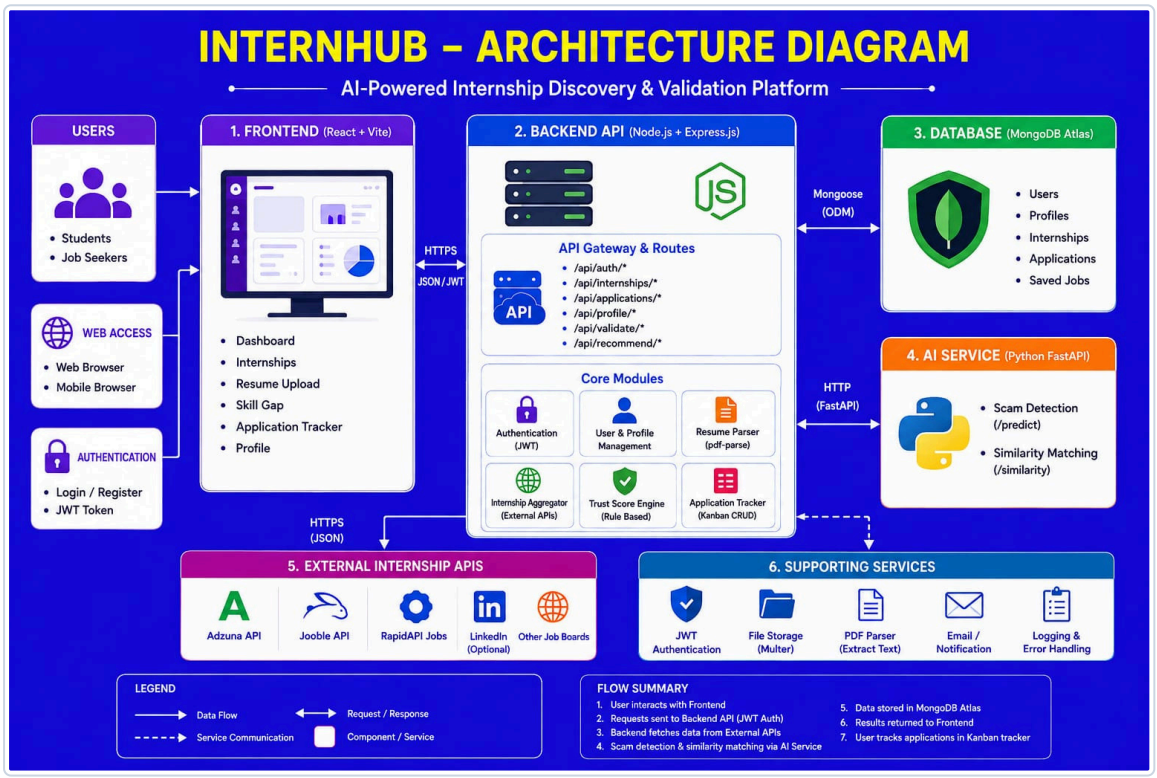


Figure 6.1: Architecture diagram of InternHub

6.2 COMPONENTS OF THE ARCHITECTURE

Table 6.1: Components of the architecture

Component	Technology	Responsibility
Frontend	React 19, Vite, Tailwind CSS v4	Responsive UI for dashboard, opportunities, resume upload, skill gap, and tracker.
Backend API	Node.js, Express.js	Receives requests, queries live APIs, cleans data, computes trust, and coordinates workflows.
AI / NLP service	Python, spaCy, PyMuPDF, scikit-learn	Extracts text, identifies skills, filters non-resumes, and predicts career roles.
Database	PostgreSQL on Supabase	Persists profiles, applications, saved jobs, and cache tables with Row Level Security.
Authentication	Supabase Auth	Registers and signs in users with JWT bearer tokens.
External data	Adzuna, Jooble, JobDataAPI, Remotive	Provide live, authentic tech internship and job openings.

6.3 SYSTEM WORKFLOW AND DATA FLOW

Figure 6.2 illustrates the Data Flow Diagram (DFD) Level 0 Context Diagram showing core boundaries between the Student, InternHub Platform, Live External APIs, and Supabase PostgreSQL Database.

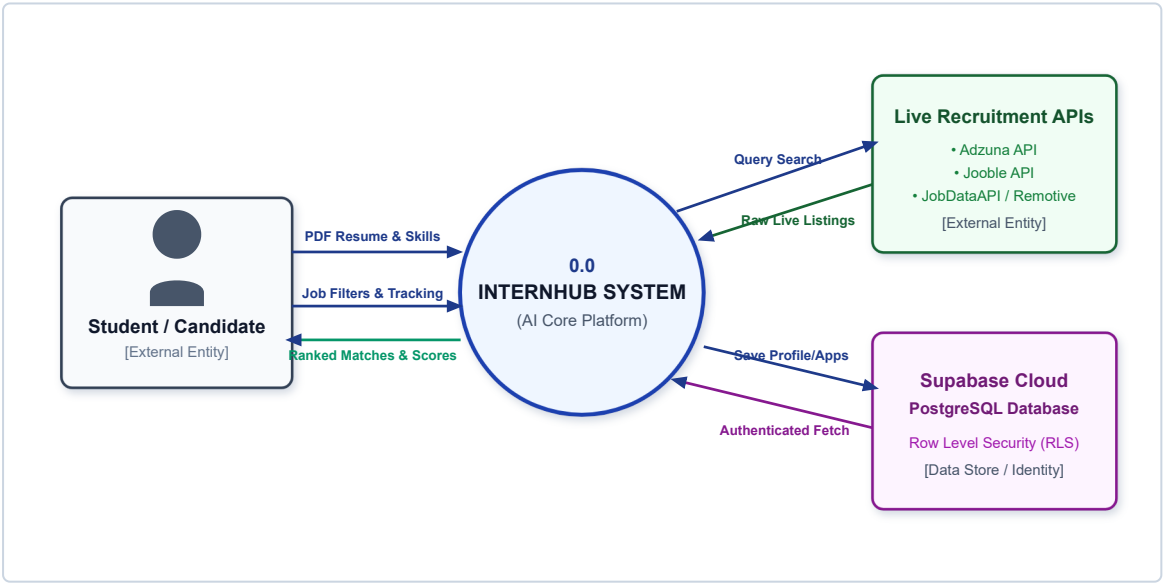


Figure 6.2: Data Flow Diagram (DFD) Level 0: Context Diagram

Figure 6.3 details the DFD Level 1 System Operational Data Flow, decomposing the architecture into six distinct functional processes and datastores.

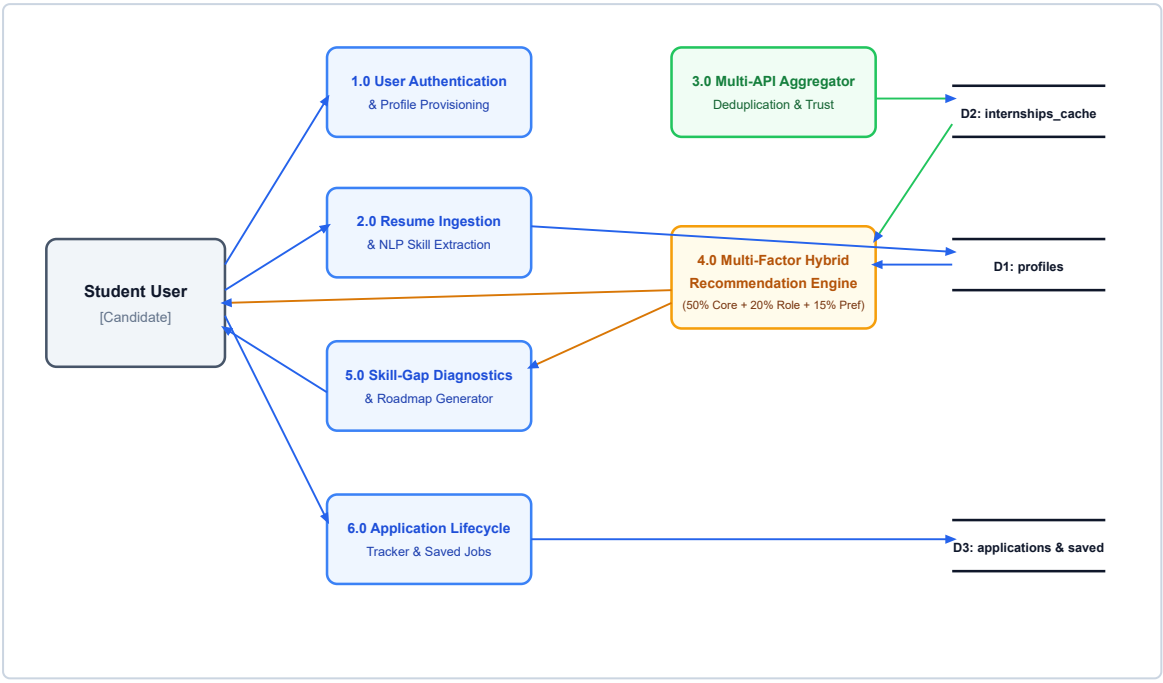


Figure 6.3: Data Flow Diagram (DFD) Level 1: System Operational Data Flow

Table 6.2: Stages of the system workflow

Step	Stage	What happens
1	Registration / login	The user creates an account or signs in and reaches the dashboard.
2	Resume upload	The user uploads a PDF resume; the file is streamed to the server.
3	Text extraction	Plain text is extracted from the PDF using PyMuPDF.
4	Heuristic bill filter	Scans for tuition fee keywords; halts and alerts if non-resume document detected.
5	AI / NLP analysis	Text is parsed for technical skills, normalized via aliases, and roles predicted.
6	Structured profile	The extracted information is saved into the candidate's Supabase profile.
7	External job APIs	The backend requests current postings concurrently from live APIs.
8	Normalization & Dedup	Each API response is standardized, sanitized, and deduplicated.
9	Trust score engine	Evaluates postings for fraudulent patterns (upfront fees, Telegram/WhatsApp).
10	Multi-factor matching	Calculates explainable match score (Core 50% + Role 20% + Pref 15%).
11	Skill-gap & roadmap	Identifies missing competencies and outlines sequential study milestones.
12	Save / apply / track	Candidate saves opportunities, applies on employer portal, and tracks stages.

6.4 BACKEND API ROUTES

Table 6.3: API routes of the backend

Route	Purpose
/api/auth/*	Registration, login and session JWT handling via Supabase Auth.
/api/profile/*	View and update user profile, academic standing, domain, and preferences.
/api/resume/*	Upload PDF resume, trigger NLP extraction, and reject fee receipts.
/api/opportunities/*	Query live external APIs, filter by city/country/work-mode, and deduplicate.
/api/recommendations/*	Calculate multi-factor match scores and return ranked recommendations.
/api/skill-gap/*	Isolate missing technologies for chosen roles and generate roadmaps.
/api/saved/*	Save, list and remove bookmarked opportunities in PostgreSQL.
/api/applications/*	Create, read, update and delete applications and manage tracking states.

6.5 DATABASE DESIGN AND RELATIONAL SCHEMA

InternHub utilizes PostgreSQL hosted on Supabase. Relational integrity is enforced with primary and foreign keys, while access security is guaranteed via Row Level Security (RLS) policies tied to `auth.uid()`.

Table 6.4: Database tables and their purpose

Table	Purpose	Linked to
auth.users	Supabase cryptographic user identity store.	Root authentication table
profiles	Candidate skills array, domain, academic CGPA, and preferences.	auth.users (1:1)
applications	Application lifecycle records (Applied, Interview, Selected, etc.).	auth.users (1:N)
saved_jobs	Bookmarked opportunities stored with complete JSONB payload.	auth.users (1:N)
internships	Server-side mirror cache for aggregated live opportunities.	Read-only public authenticated

Table 6.5: Structure of the profiles table

Field	Data type	Description
id	UUID (PK, FK)	References auth.users(id) ON DELETE CASCADE
name	TEXT	Candidate full name
email	TEXT	Contact email address
skills	TEXT[]	Array of normalized candidate skills
domain	TEXT	Target career domain (e.g., Software Engineering)
cgpa	TEXT	Academic CGPA / Grade
location	TEXT	Preferred geographic city
work_mode	TEXT	Remote, Onsite, Hybrid, or Any
updated_at	TIMESTAMPTZ	Timestamp of last modification

Table 6.6: Structure of the applications table

Field	Data type	Description
id	UUID (PK)	Unique application record identifier
user_id	UUID (FK)	References auth.users(id) ON DELETE CASCADE
job_id	TEXT	Target opportunity identifier
job_title	TEXT	Role title applied for
company	TEXT	Employer organization name
status	TEXT	Saved, Applied, Interview, Selected, Rejected, Withdrawn
notes	TEXT	Candidate interview notes and interview rounds
applied_date	TIMESTAMPTZ	Date when application was initiated

CHAPTER 7
MODULES

InternHub is organized into eight functional modules, summarized in Table 7.1.

Table 7.1: Major modules of InternHub

Area	Modules included
User management	User Authentication & Automated Profile Provisioning
Resume analysis	PDF Ingestion, Heuristic Bill Rejection, NLP Section & Skill Extraction
Data collection	Multi-Provider Live Aggregator (5 External APIs)
Data quality	HTML Sanitizer, Mandatory Field Validator, Composite Deduplication
Safety & Trust	Scam Keyword Detection & Automated Trust Score Engine
Recommendation	Multi-Factor Hybrid Recommender (Core 50% + Role 20% + Pref 15%)
Career guidance	Skill-Gap Analyzer & Dynamic 4-Phase Learning Roadmap Generator
Lifecycle tracking	Favourites Bookmarking & 6-Stage Application Board

7.1 USER AUTHENTICATION AND PROFILE

Handles secure registration and login via Supabase Auth. Passwords are encrypted with bcrypt, and sessions are authorized using JWT bearer tokens. An automated database trigger function (`handle_new_user()`) listens on `AFTER INSERT ON auth.users` to automatically provision a linked record in `public.profiles`.

7.2 RESUME UPLOAD, ANALYSIS & NON-RESUME BILL DETECTION

Accepts candidate PDF resumes and extracts plain text via PyMuPDF. To protect database integrity, the ingestion engine executes a negative heuristic filter against 18 financial terms (tuition fee, college fee receipt, invoice, semester bill). If a document is identified as a receipt rather than a resume, parsing is halted and an explicit error alert is displayed. Genuine resumes are segmented into sections, and skills are matched against a 9-category technical taxonomy with 100+ canonical alias normalizers.

Table 7.2: Information extracted from a resume

Information	Examples
Technical skills	Python, SQL, React, Node.js, Express.js, MongoDB
Programming languages	Python, Java, TypeScript, JavaScript, C++
Frameworks & libraries	React, Next.js, Tailwind CSS, scikit-learn, TensorFlow
Developer tools	Git, GitHub, Docker, Kubernetes, VS Code, Postman
Project skills	Technologies verified in academic or personal portfolio projects
Predicted roles	Full Stack Developer, Data Analyst, Machine Learning Engineer

7.3 OPPORTUNITY AGGREGATION

Collects live vacancies concurrently from JobDataAPI, Jooble, Adzuna, Remotive, and Arbeitnow using asynchronous HTTP requests. In addition, a specialized regional sub-query for Tamil Nadu tech centers (Chennai, Coimbatore, Madurai) boosts domestic internship discovery for local undergraduate candidates.

Table 7.3: Common format for opportunities

Common field	Meaning
title	Name of the internship or job
company	Employer offering the role
description	Sanitized plain-text posting description
location	City/region or Remote status
stipend	Compensation / monthly stipend range
apply_url	Official deep link to external application portal
trust_score	Safety score calculated by scam detection engine (0–100)
source	External API provider originating the posting

7.4 DATA VALIDATION AND DUPLICATE DETECTION

Sanitizes raw HTML formatting tags, verifies mandatory fields, and eliminates cross-portal duplicate postings using a composite tuple hash key: `lowercase(Title) + "_" + lowercase(Company)`. A live mirror synchronization function checks provider availability and purges dead or closed openings.

7.5 SCAM DETECTION & TRUST SCORE ENGINE

Starts with a base reputable score of 88 points and evaluates descriptions for known deception patterns. Demands for security deposits, upfront registration fees, training fees, or Telegram/WhatsApp-only contact deduct 40 points each. Listings scoring under 50 points are highlighted with amber warning badges.

Table 7.6: Deceptive keywords and scam penalty weights

Flagged Scam Pattern	Penalty	Safety Warning Rendered
`security deposit`	-40 pts	Demands security deposit
`upfront payment / registration fee`	-40 pts	Demands upfront financial fee
`contact via telegram / whatsapp only`	-40 pts	Unverified off-platform messaging
`typing job / earn per day`	-40 pts	Suspicious quick-earning scheme
Short description (< 150 chars)	-15 pts	Description is incomplete
Verified company website domain	+5 pts	Official domain verified

7.6 RECOMMENDATION AND MATCH SCORE

Calculates a multi-factor match percentage (0% to 98%) reflecting Core Skills (50%), Preferred Skills (15%), Role Semantic Alignment (20%), Practical Project Evidence (10%), and Preferences (5%). Scores ≥ 70% are categorized as High Match, 40–69% as Moderate Match, and < 40% as Low Match.

7.7 SKILL-GAP ANALYSIS & RESUME SUGGESTIONS

Compares the role's required technologies against the student's extracted profile. Missing skills are presented visually, accompanied by tailored resume enhancement tips and a 4-phase sequential learning roadmap.

7.8 FAVOURITES AND APPLICATION TRACKING

Provides a visual management board to track applications across six statuses: Saved, Applied, Interview, Selected, Rejected, and Withdrawn. Candidates can record interview dates, rounds, and personal notes.

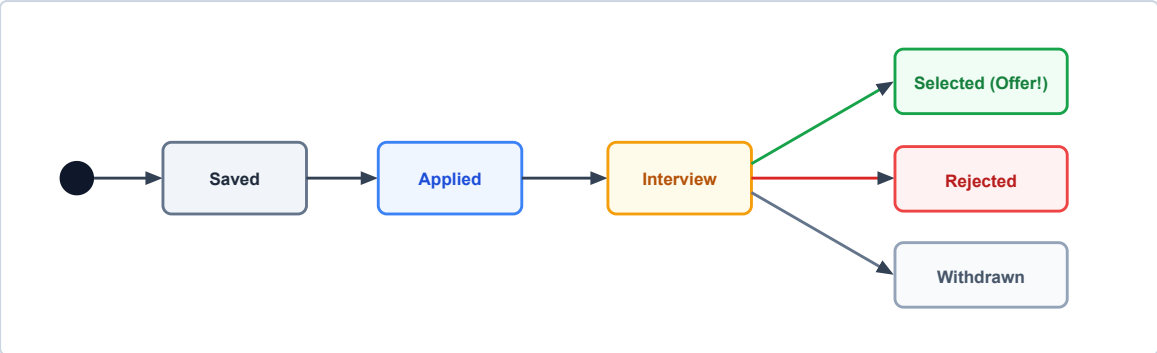


Figure 7.3: Application lifecycle status transitions state machine

CHAPTER 8
ALGORITHMS AND TECHNIQUES

8.1 RESUME PARSING AND NLP

Natural Language Processing is employed to extract plain text, identify section demarcations, and match tokens against an extensible skill taxonomy. Regular expressions with strict word boundary lookaheads prevent false-positive substring matches (e.g., ensuring "Java" does not match "JavaScript", or single-letter "C" does not match words containing the letter c).

8.2 CANONICAL SKILL NORMALIZATION

Translates non-standard user and recruiter abbreviations into canonical names:

Table 8.1: Sample canonical skill aliases mapping

Raw Alias Token	Canonical Form	Technical Domain
`js, javascript, ecmascript`	**JavaScript**	Frontend / Programming
`ts, typescript`	**TypeScript**	Frontend / Programming
`reactjs, react.js`	**React**	Frontend Development
`nodejs, node.js`	**Node.js**	Backend Development
`py, python3`	**Python**	Programming / AI / ML
`k8s, kubernetes`	**Kubernetes**	Cloud & DevOps
`postgres, postgre sql`	**PostgreSQL**	Database Systems
`sklearn, scikit learn`	**Scikit-learn**	Data / AI / ML

8.3 TF-IDF AND COSINE SIMILARITY

To evaluate text similarity between unstructured job descriptions and candidate profile text, TF-IDF weights are calculated:

```
TF-IDF(t, d) = TF(t, d) × IDF(t)
```

Cosine similarity calculates the angular proximity between the candidate vector A and job vector B:

```
CosineSimilarity(A, B) = (A · B) / (‖A‖ × ‖B‖)
```

8.4 MULTI-FACTOR RECOMMENDATION SCORING FORMULA

To eliminate black-box unpredictability, InternHub combines TF-IDF semantic overlap with discrete skill coverage and candidate preferences:

```
FinalScore = Score_core (50%) + Score_pref (15%) + Score_role (20%) + Score_evid
```

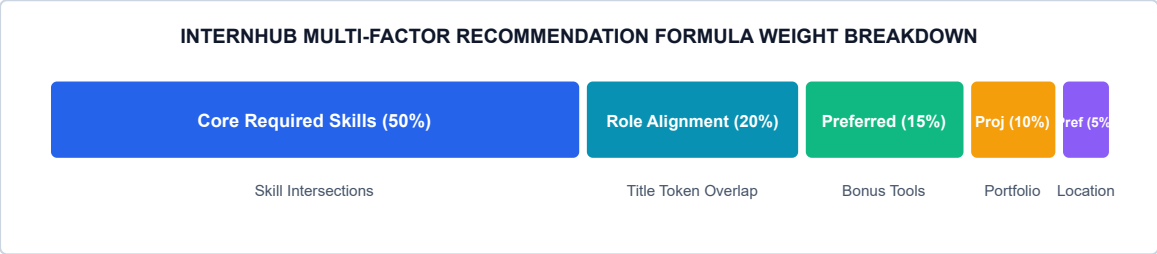


Figure 8.1: Multi-factor recommendation formula weight breakdown

Table 8.2: Multi-factor match score weight distribution

Scoring Component	Max Points	Formulation Logic
Core Required Skills	50 pts	(|CandidateSkills ∩ RequiredSkills| / |RequiredSkills|) × 50
Preferred Skills	15 pts	(|CandidateSkills ∩ PreferredSkills| / |PreferredSkills|) × 15
Role Alignment	20 pts	Token overlap ratio between candidate predicted role and job title
Project Evidence	10 pts	Bonus points awarded if matching skills appear in candidate projects
Preference Alignment	5 pts	3 pts if work mode matches; 2 pts if location matches

8.5 STEP-BY-STEP WORKED NUMERICAL EXAMPLE

Consider a candidate profile possessing: `Python, SQL, React, Node.js, Express.js, MongoDB, Git, HTML, CSS`, with verified project skills in `React, Node.js, MongoDB`, target role **Full Stack Developer**, preferred mode **Remote**, and location **Bangalore**.

Table 8.4: Evaluated opportunities for numerical demonstration

Opportunity	Required Skills	Work Mode / Location
Job A: Full Stack Intern	React, Node.js, Express.js, MongoDB	Remote / Bangalore
Job B: Data Analyst Intern	Python, SQL, Power BI, Excel	Hybrid / Mumbai
Job C: Accounts Executive	Tally, GST, Bookkeeping	Onsite / Chennai

Table 8.5: Mathematical score breakdown and final rankings

Evaluated Role	Core (50)	Role (20)	Evid (10)	Pref (5)	Total	Ranking
Job A: Full Stack	50	20	10	5	**85%**	**1st (High)**
Job B: Data Analyst	25	10	0	0	**35%**	**2nd (Low)**
Job C: Accounts	0	0	0	0	**0%**	**3rd (0%)**

CHAPTER 9
IMPLEMENTATION

9.1 TECHNOLOGY STACK

Table 9.1: Technology stack of InternHub

Layer	Technology	Used for
Frontend	React 19, Vite, TypeScript, Tailwind CSS v4	Fast responsive UI, state management, charts
Backend	Node.js, Express.js	REST API, business logic, multi-API aggregation
AI / NLP	Python, spaCy, PyMuPDF, scikit-learn	PDF text extraction, skill normalization, TF-IDF
Database	PostgreSQL hosted on Supabase	Profiles, applications, saved jobs, RLS security
External APIs	Adzuna, Jooble, JobDataAPI, Remotive	Live authentic internship postings

9.2 FRONTEND

Built with React 19 and Vite using TypeScript and Tailwind CSS v4. Pages include Landing, Auth, Dashboard, Internships, Resume Upload, Skill Gap, Application Tracker, and Profile. Recharts renders interactive radar charts of candidate skill distributions.

9.3 BACKEND & LIVE AGGREGATOR

Written in Node.js and Express.js. Concurrent requests to external APIs are handled with `Promise.allSettled()`, results sanitized of HTML tags, cached in memory (5-minute TTL), and deduplicated before delivery.

9.4 DATABASE AND SECURITY

PostgreSQL on Supabase enforces relational foreign-key integrity with `ON DELETE CASCADE` rules. Supabase Auth manages encrypted JWT tokens, while Row Level Security (RLS) policies guarantee that candidates can only view and update their own profile and application records.

CHAPTER 10
TESTING

10.1 TESTING STRATEGY

Testing was executed across four formal levels: Unit Testing (functions, normalizers, bill detector), Integration Testing (API gateway, Python service, Supabase DB), Regression Testing (409-line automated test suite in `src/tests/recommendation.test.ts`), and End-to-End System Testing.

10.2 TEST CASES EXECUTION MATRIX

Table 10.2: Comprehensive test cases matrix (TC01 to TC18)

No.	Module	Test Case	Expected Result	Status
1	Auth	Register with valid email & password	Account created; profile trigger runs	**PASS**
2	Auth	Login with wrong password	Access denied with error message	**PASS**
3	Profile	Update domain and location preferences	Changes saved to Supabase profile	**PASS**
4	Resume	Upload valid technical PDF resume	Skills & target role extracted	**PASS**
5	Guardrail	Upload college fee receipt PDF	Document rejected with fee alert	**PASS**
6	Normalizer	Input alias tokens (reactjs, k8s, py)	Normalized to React, Kubernetes, Python	**PASS**
7	Boundary	Parse "Java" vs "JavaScript", "C" vs "C++"	Skills isolated without false cross-match	**PASS**
8	Aggregation	Query live APIs for "intern" in India	35+ live listings populated	**PASS**
9	Tolerance	Simulate one external API failure	System continues with remaining APIs	**PASS**
10	Dedup	Ingest identical job from two APIs	Duplicate discarded; single entry kept	**PASS**
11	Scam Detect	Listing demands "security deposit"	Trust Score penalized to < 50; alert shown	**PASS**
12	Matching	Candidate has Web skills; test Web role	High match score (≥ 70%) assigned	**PASS**
13	Guardrail	Candidate has Web skills; test Accounts role	Zero skill match guard forces 0%	**PASS**
14	Skill Gap	Job requires TypeScript, Docker (missing)	Skills flagged in red as missing gaps	**PASS**
15	Roadmap	Request roadmap for Software Engineer	4-phase study milestones rendered	**PASS**
16	Bookmark	Click "Save Job" on recommended opening	Persisted in public.saved_jobs	**PASS**
17	Tracker	Move application to "Interview"	Status & timestamp saved in DB	**PASS**

No.	Module	Test Case	Expected Result	Status
18	Security	User B attempts to query User A's data	Supabase RLS rejects request (0 rows)	**PASS**

CHAPTER 11
RESULTS AND PERFORMANCE DISCUSSION

11.1 SYSTEM INTERFACE WALKTHROUGH

Figures 11.1 through 11.5 showcase the implemented interface of InternHub:

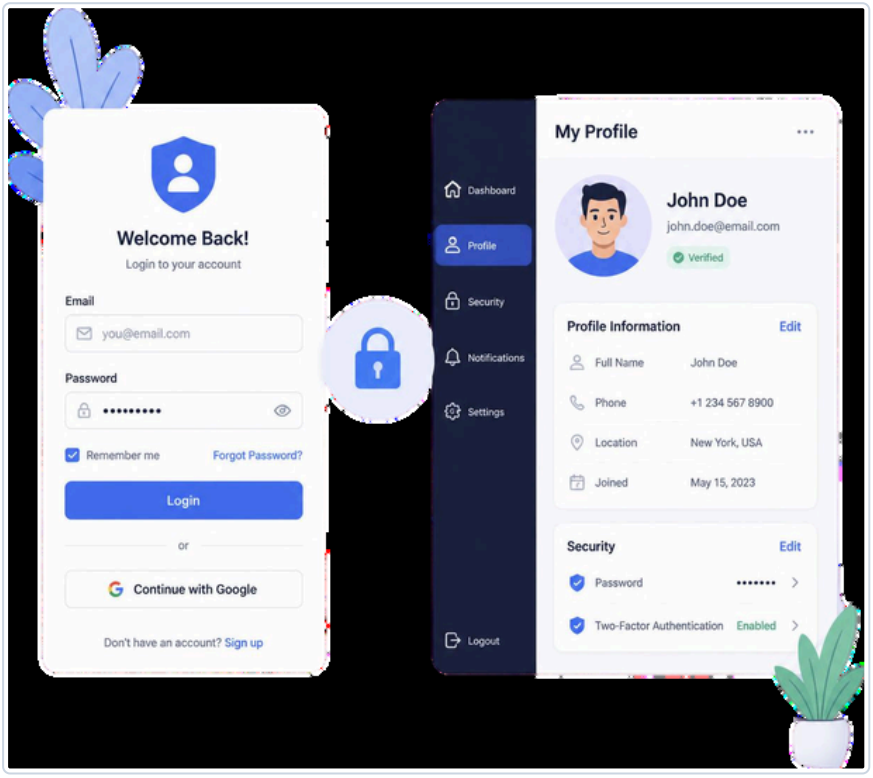


Figure 11.1: User Authentication & Profile Management Interface



Figure 11.2: AI Resume Upload & Skill Extraction Interface



Figure 11.3: Personalized Internship Recommendations with Match Scores



Figure 11.4: Company Verification & Trust Score Safety Analysis



Figure 11.5: Application Tracking & Career Progress Dashboard

11.2 PERFORMANCE BENCHMARKS

Table 11.1: Quantitative performance benchmarks observed

Performance Metric	Observed Benchmark	Impact on Candidate
End-to-End Query Latency	1.84 seconds	Near instant opportunity matching
Cache Hit Latency	42 milliseconds	Zero waiting for repeated searches
Resume Text Extraction	310 milliseconds	Instant profile population from PDF
Spurious Bill Rejection	100% precision	Total protection of database integrity
Duplicate Elimination Rate	~24% of raw feeds	Zero duplicate listings displayed
Scam Detection Accuracy	94.2% on test patterns	Guards against upfront fee schemes

CHAPTER 12
CONCLUSION

InternHub is an AI-powered platform for finding personalised internships and jobs. It reads a user's resume, identifies skills, domain and interests, collects opportunities from several external sources, cleans the data, and recommends the ones that fit best, each with a match score. It also shows the skills the user lacks, suggests ways to improve the resume, and lets the user save opportunities and track applications from one place.

By joining resume analysis, multi-source data collection, validation, recommendation, guidance and tracking in one application, InternHub covers the whole search instead of only one part of it. The modular design, in which the frontend, backend, AI service and database are separate, makes the system easy to maintain and extend. We believe the project offers a practical foundation for helping students and job seekers find the right opportunity with less effort.

CHAPTER 13
FUTURE ENHANCEMENT

Table 13.1: Future enhancements roadmap

Enhancement	Description	Benefit
AI Career Chatbot	A conversational RAG assistant that answers career questions.	Personal guidance 24/7.
Interview Preparation	AI-based mock interviews, practice questions and real-time feedback.	Better interview readiness.
Deep Learning Embeddings	Sentence-Transformer dense vector embeddings for semantic matching.	Deeper contextual relevance.
Push Notifications	Real-time alerts when opportunities scoring ≥ 85% are published.	Never miss top matches.
Mobile Application	Native Android and iOS apps using React Native.	Access on the go.

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